

Lesson **Two**

Project Management & Process

Effective Project Management: Traditional, Agile, Extreme

Ch02: What Is Project Management?

Ch03: What Is Strategic Project Management?

***Presented by
Bowen DU***

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Definition of Project Management

Project management is an organized common-sense approach that utilizes the appropriate client involvement in order to meet sponsor needs and deliver the expected incremental business value.

By Wysocki

The application of knowledge, skills, tools and techniques to project activities to meet the project requirements.

By PMI

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The basic questions of Project Management

Project management should answer the following **six** questions:

- What business is being addressed by this project?
- What do you need to do?
- What will you do?
- How will you do it?
- How will you know you did it?
- How well did you do?

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Challenges to Effective Project Management

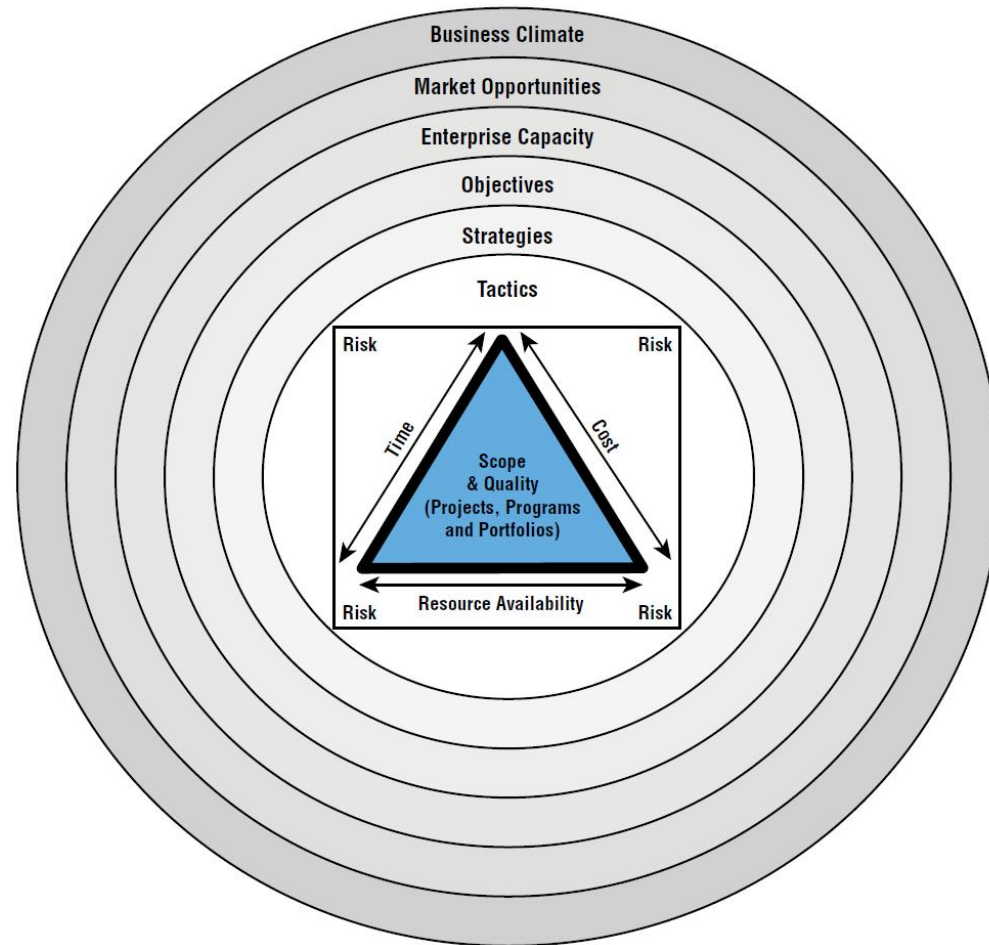
- Flexibility and Adaptability
- Deep Understanding of the Business and Its Systems
- Take Charge of the Project and Its Management

Always remember

Project management is organized common sense.
If it doesn't make sense, don't do it.

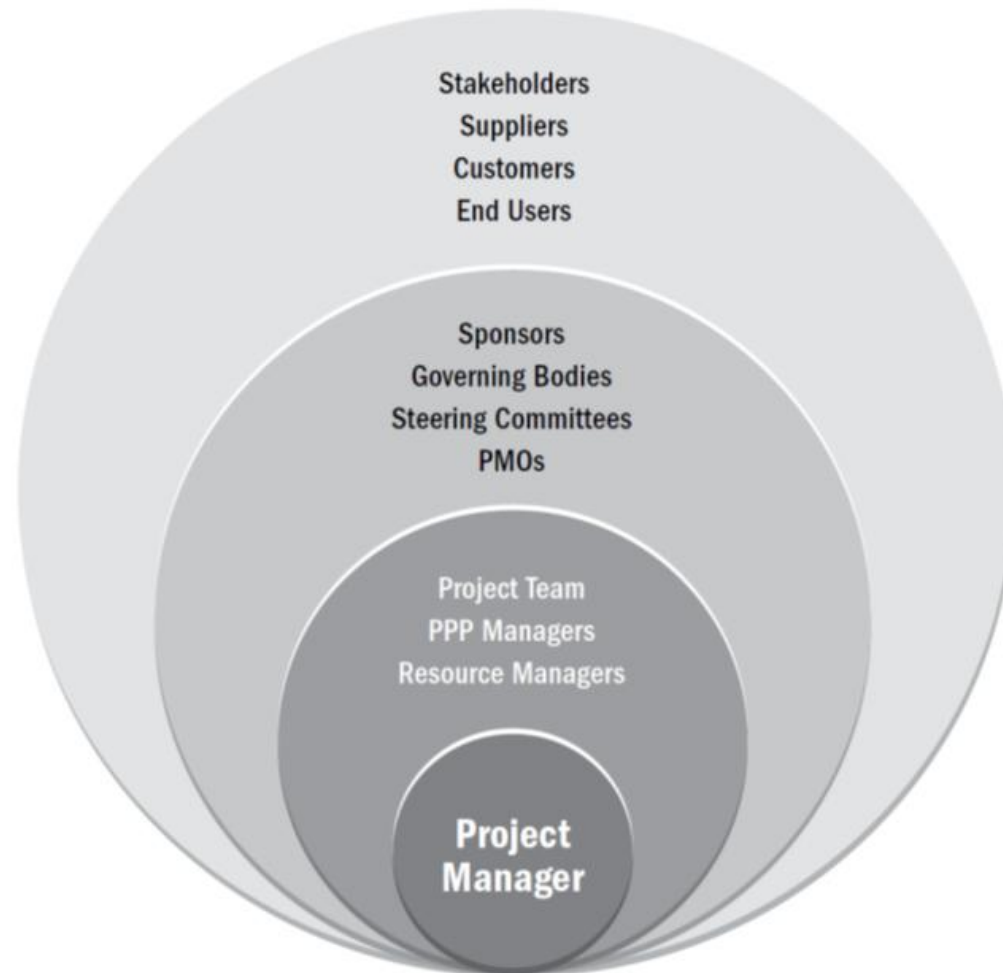
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Business Environment



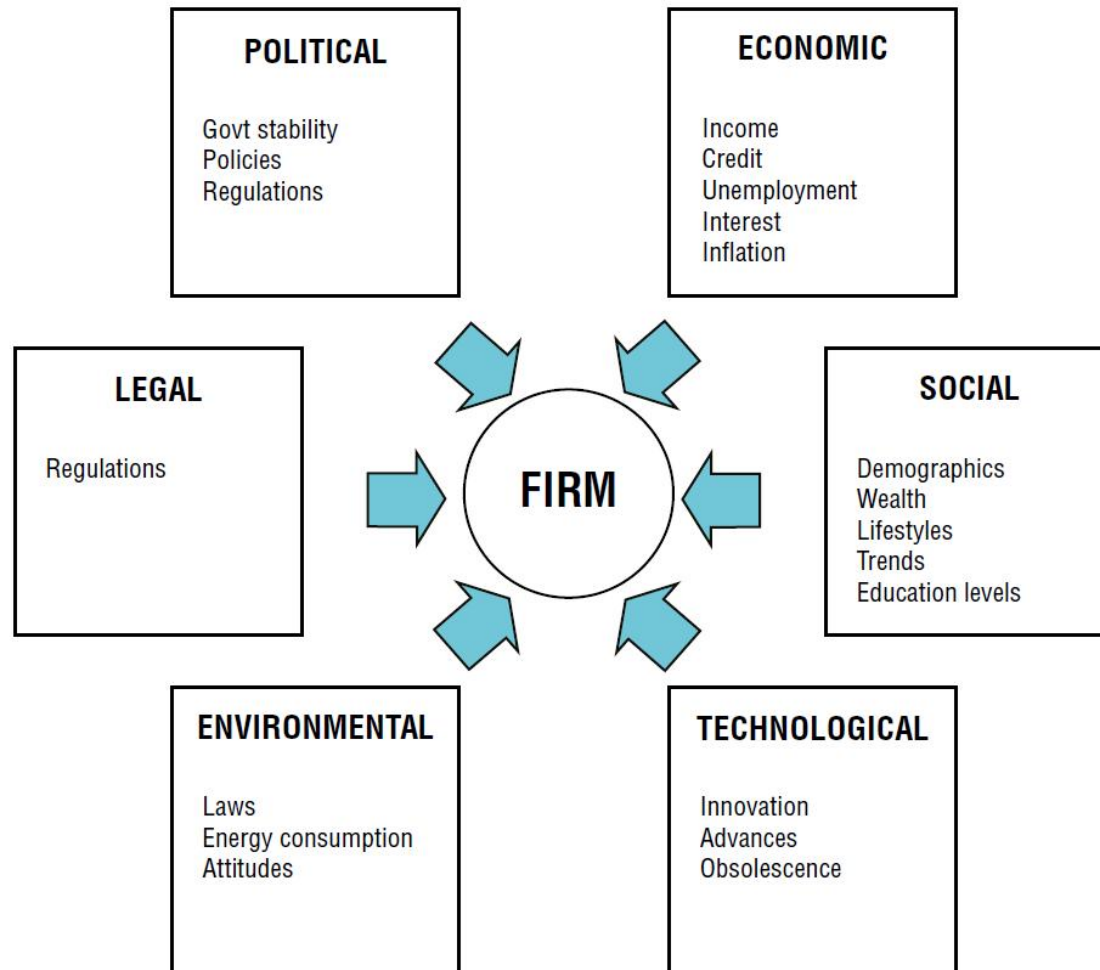
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Stakeholders



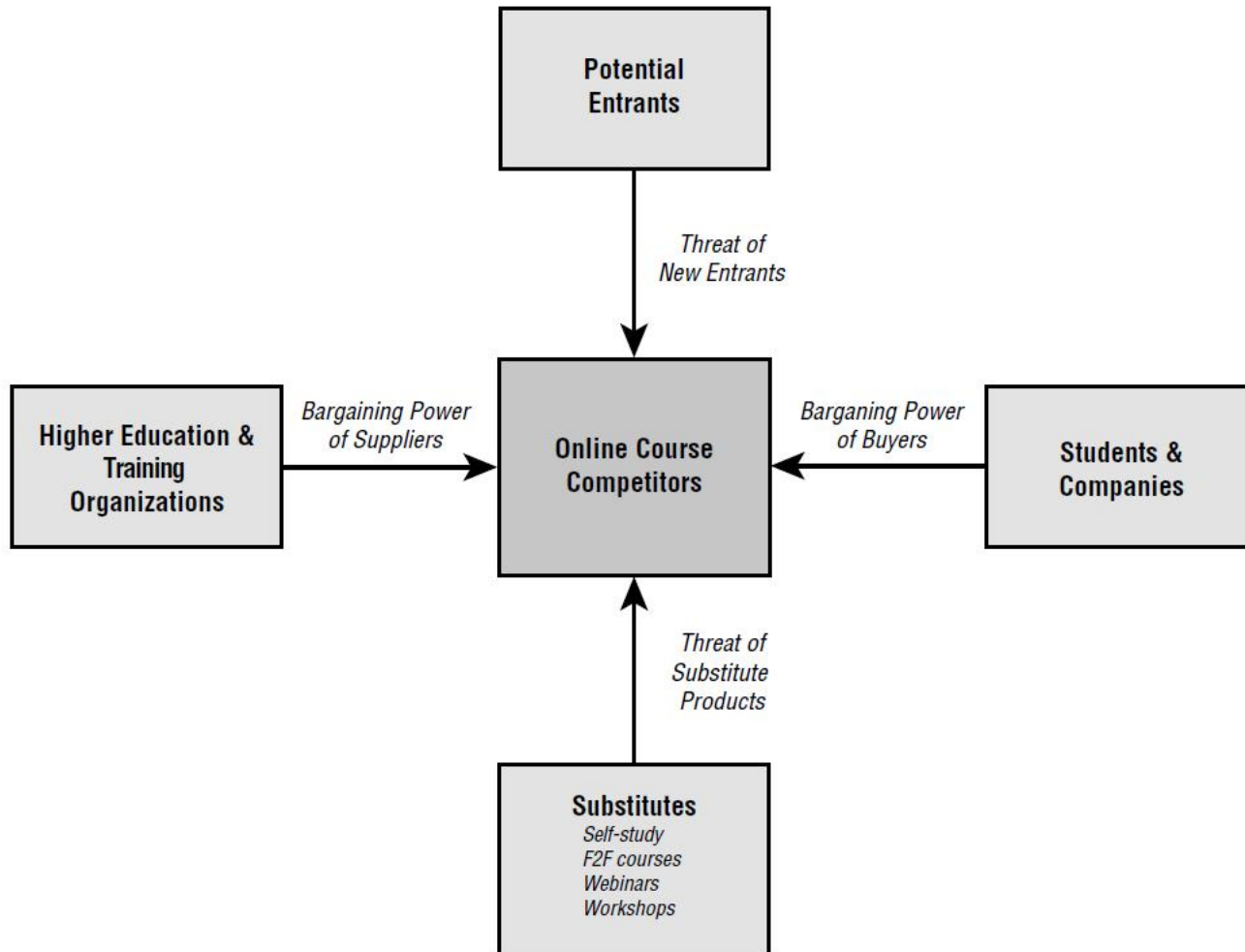
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PESTEL Model



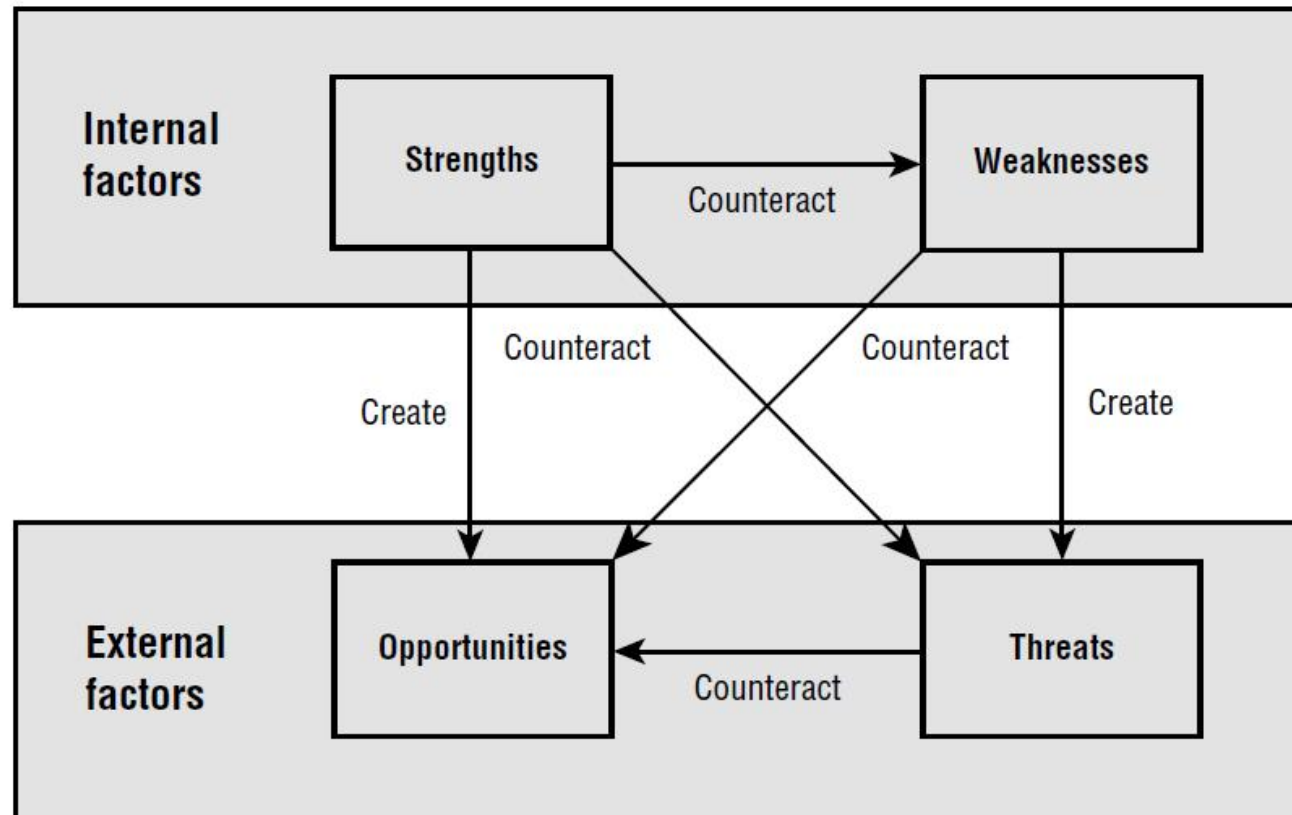
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Porter's Competitive Forces Applied to the Case Study



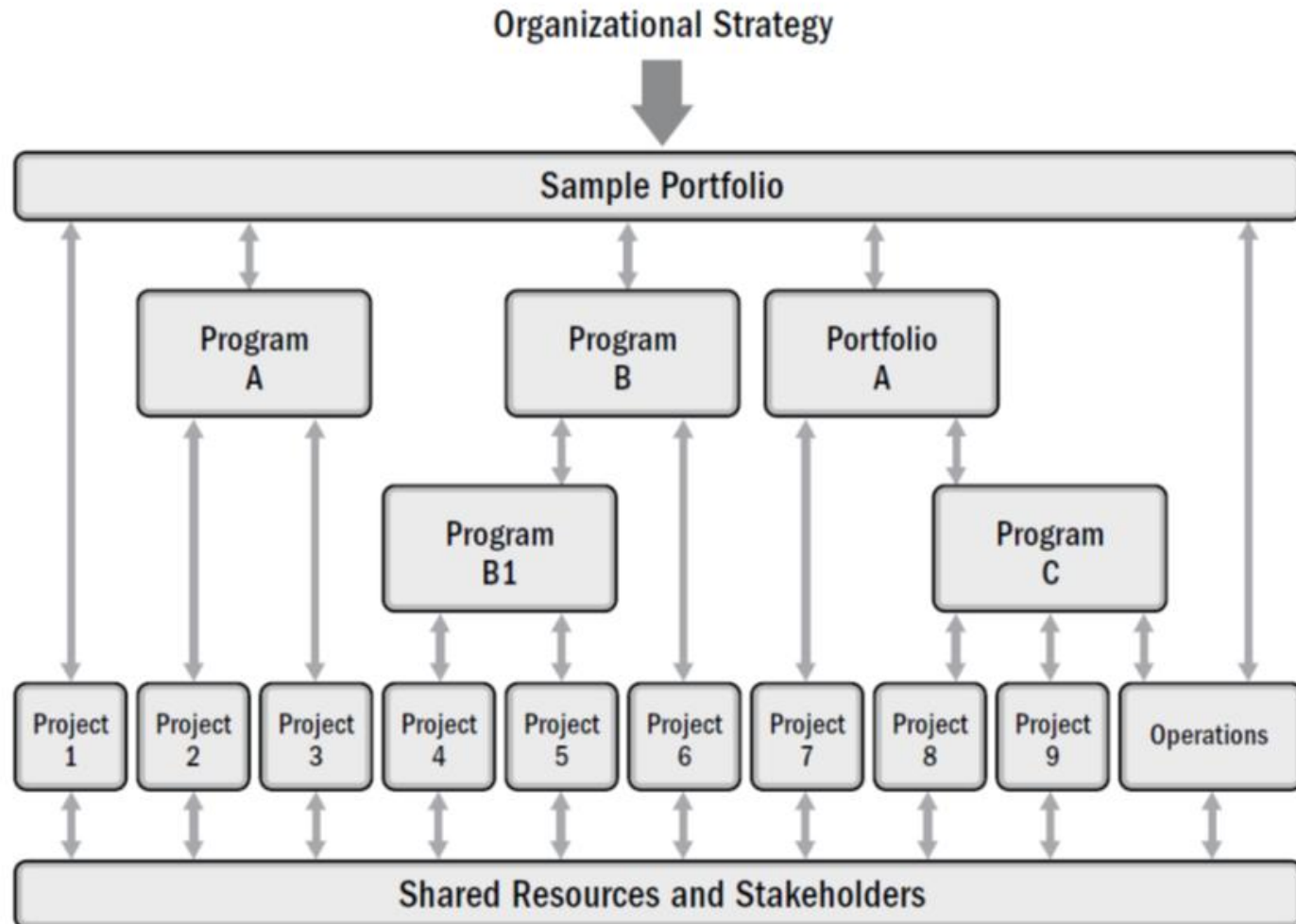
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SWOT Model



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Portfolio, Program & Project



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Managing the Creeps(蠕变)

- Scope Creep
- Hope Creep
- Effort Creep
- Feature Creep

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Managing the Creeps(蠕变)

■ Exercise 1# in-class

《Mythical man-month》 by Frederick Brooks

Case study: Regenerative Schedule Disaster

(Page 21-26)

What is your thinking about this case?

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Definition of Requirements (IIBA BABOK, 2009)

A requirement is:

1. A condition or capability needed by a **stakeholder** to solve a problem or achieve an objective.
2. A condition or capability that must be met or possessed by a **solution** or **solution component** to satisfy a contract, standard, specification, or other formally imposed documents.
3. A **documented representation** of a condition or capability as in (1) or (2).

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A Business Value Definition of Requirements

A requirement is a desired end-state whose successful integration into the solution meets one or more needs and delivers **specific, measurable** and **incremental business value** to the organization.

The set of requirements forms a necessary and sufficient set for achieving expected business value.

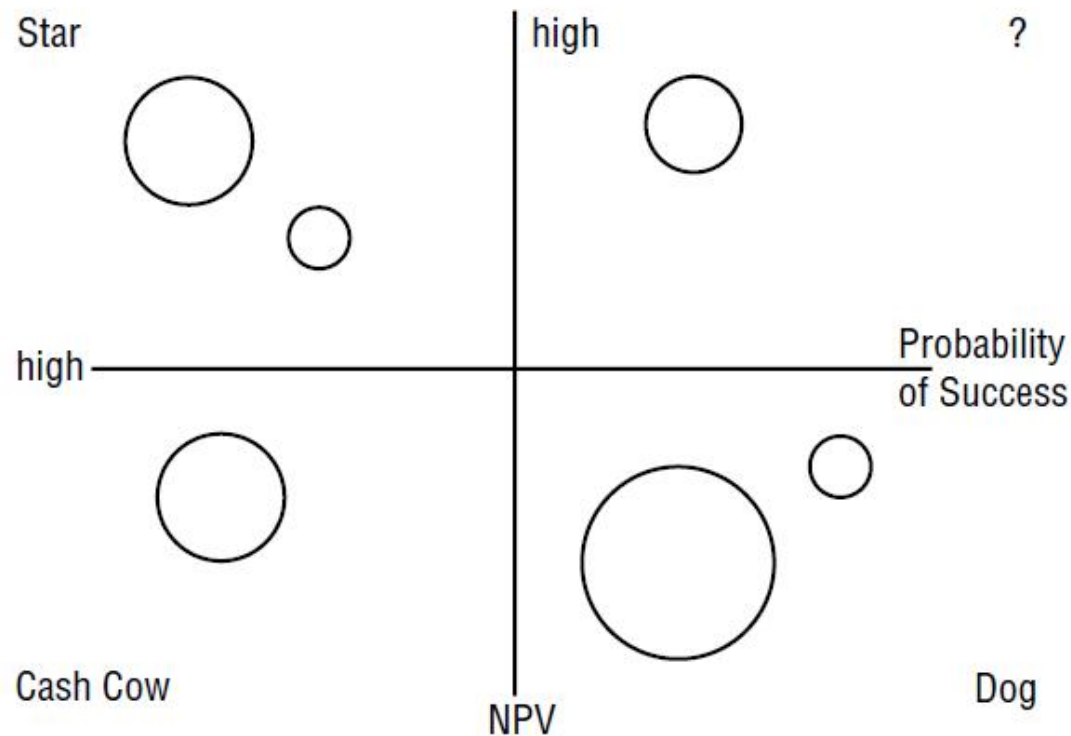
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The Value of The Definition of Requirements

- Reduces the number of requirements from dozens (even hundreds) to 6 or 8.
- Identifying the complete definition of most requirements happens only through iteration.
- Simplifies the search for a solution with acceptable business value.
- Choosing among alternative solution directions is simplified.
- Provides for better use of scarce resources (money, time, and people).
- A working(有效的) definition.

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Market Opportunities



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Generating Complete Requirements Is Not Likely

- Changing market conditions
- Actions of competitors
- Technology advances
- Client discovery
- Changing priorities

Exercise in-class:

Please give some **examples** for the above scenario.

You can never know for sure that requirements are complete.
When in doubt, err on the side that they are not complete.

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Definition of a Project Management Life Cycle

A project management life cycle is a sequence of processes that includes:

- Scoping
- Planning
- Launching
- Monitoring & Controlling
- Closing

NOTE: Every valid project management approach must include these processes.

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Introducing Project Management Life Cycles (PMLC)

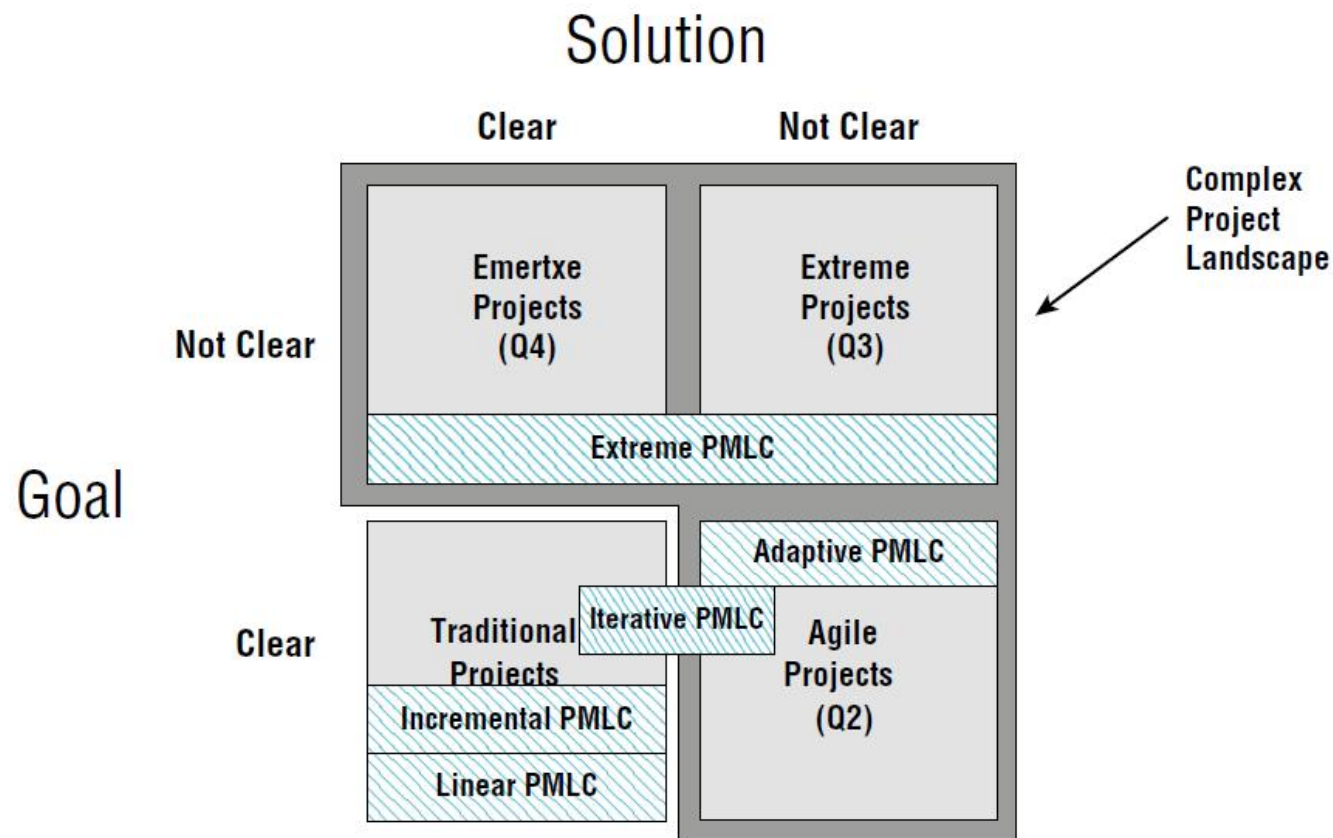


Figure 2.1: Five PMLC types

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TPM – Traditional Project Management

SOLUTION			
		Clear	Not Clear
GOAL	Not Clear		
	Clear	TPM	

Characteristics

- Low Complexity
- Few Scope Change Requests
- Well-Understood Technology Infrastructure
- Low Risk
- Experienced and Skilled Project Teams
- Plan-driven TPM Projects

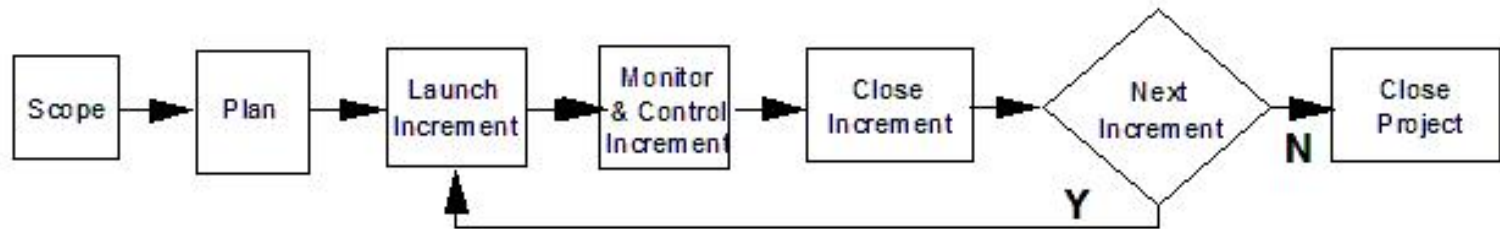
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Linear PMLC Model



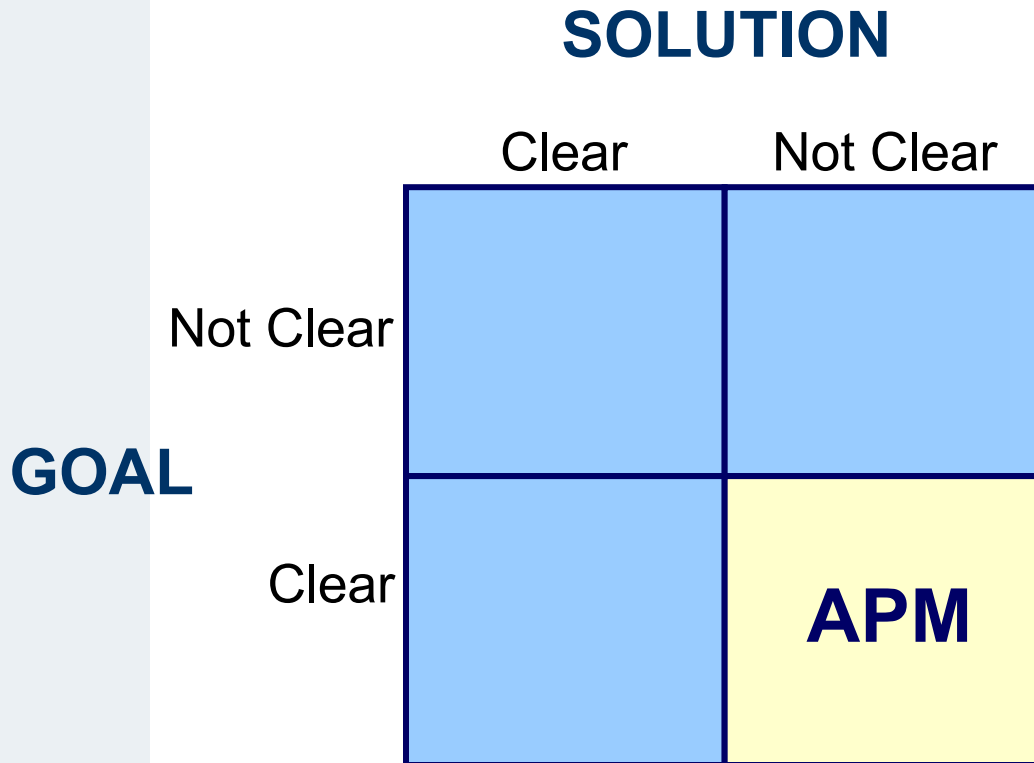
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Incremental PMLC Model



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APM – Agile Project Management

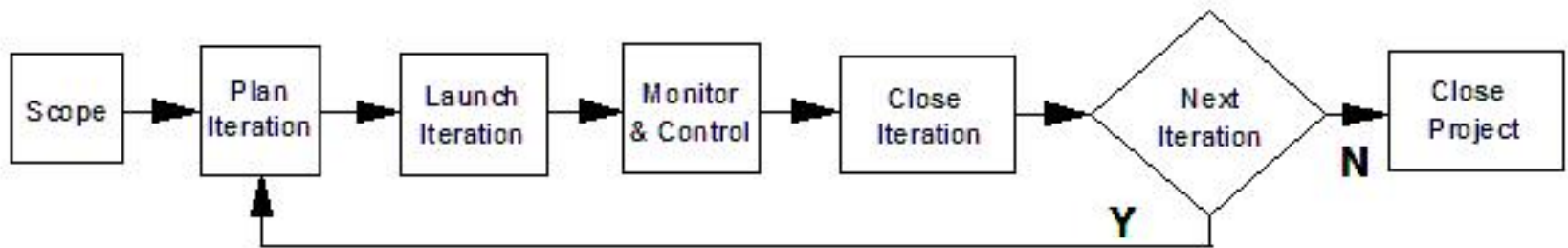


Characteristics

- A Critical Problem without a Known Solution
- A Previously Untapped Business Opportunity
- Change-driven APM Projects
- Critical to the Organization
- Meaningful Client Involvement is Essential
- Use Small Co-located Teams

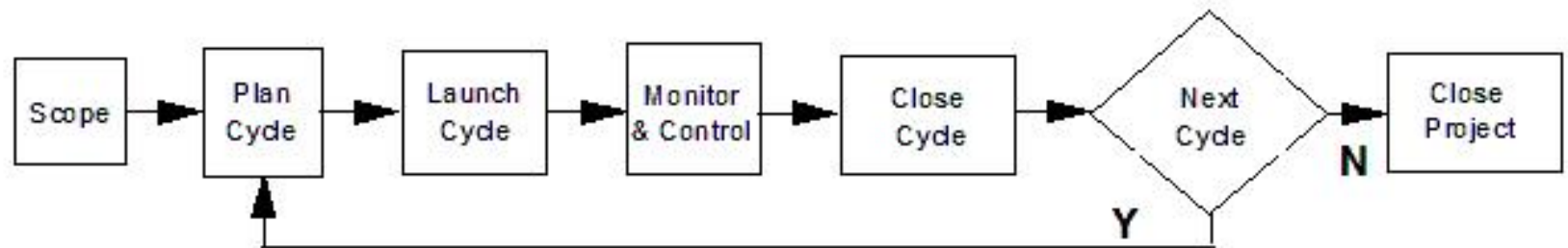
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Iterative PMLC Model



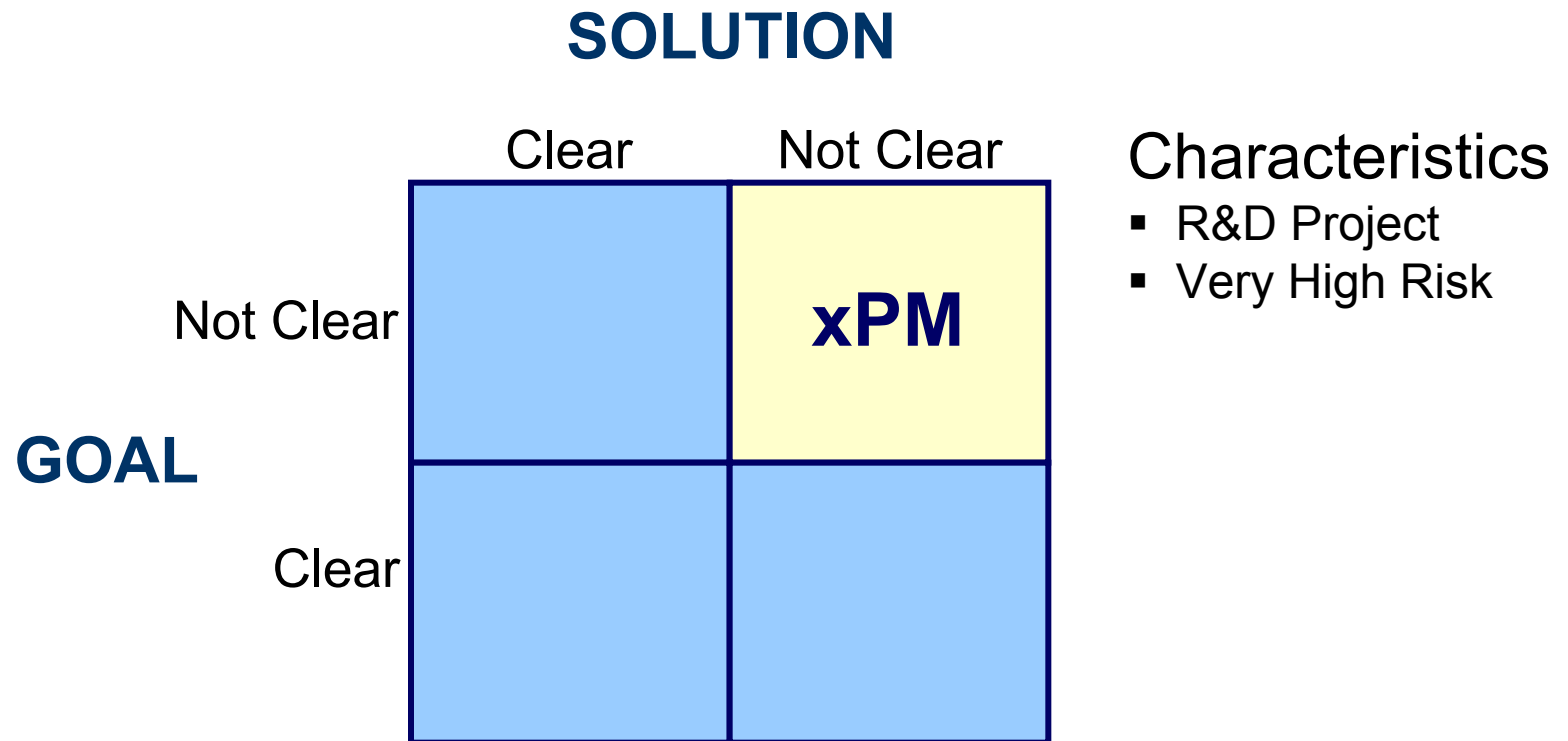
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Adaptive PMLC Model



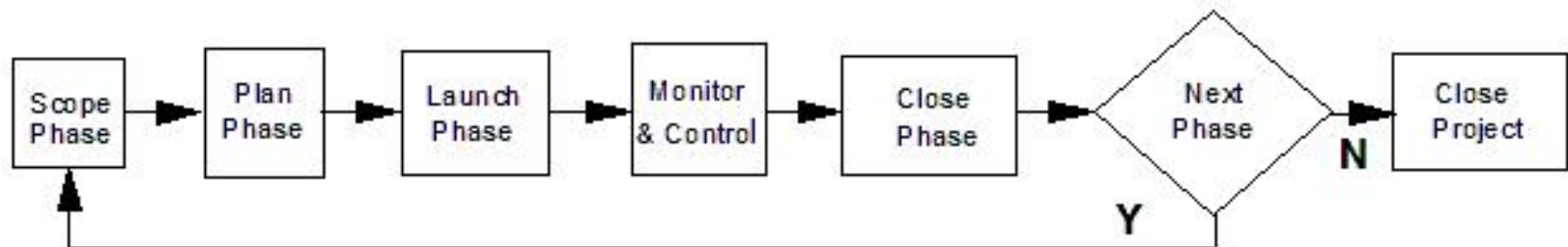
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xPM – Extreme Project Management



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Extreme PMLC Model



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Emertxe Project Management

		SOLUTION	
		Clear	Not Clear
GOAL	Not Clear	MPx	
	Clear		

Characteristics

- A New Technology without a Known Application
- A Solution out Looking for a Problem to Solve

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Examples of Each Project Type

TPM Install an intranet network in field office

APF Put a man on the moon by the end of the decade and return him safely

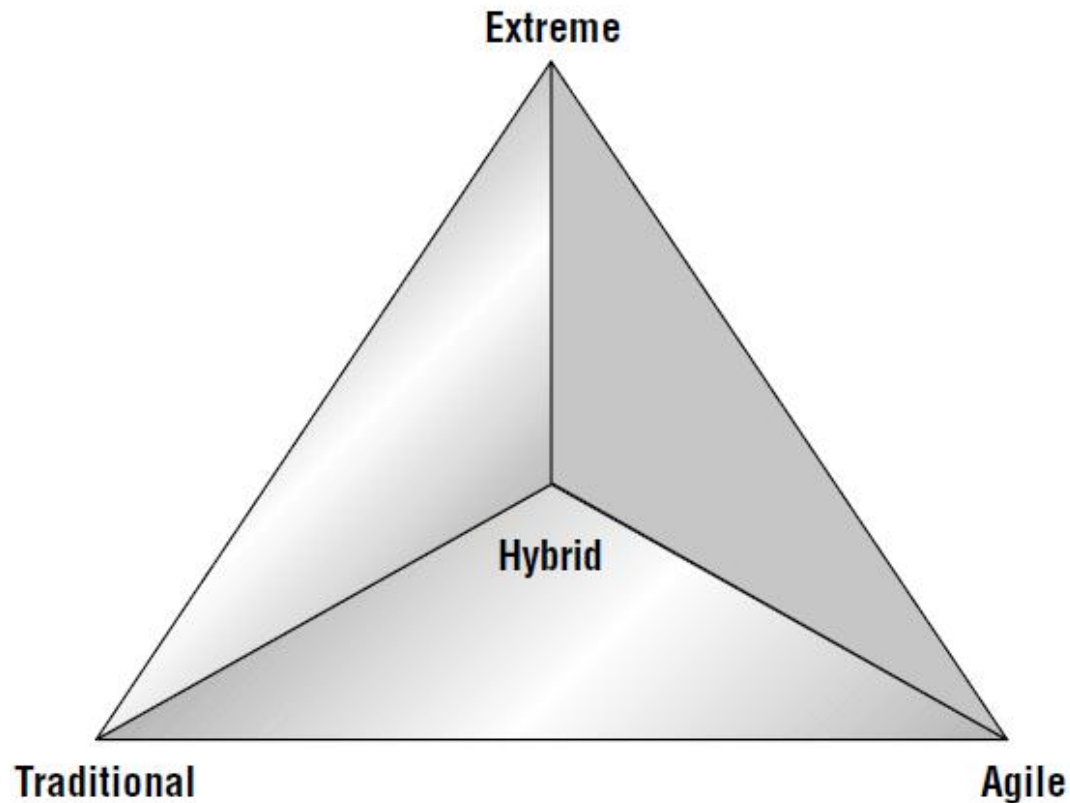
xPM Cure the common cold

MPx Give an example

		SOLUTION	
		Clear	Not Clear
GOAL	Not Clear	MPx	xPM
	Clear	TPM	APM

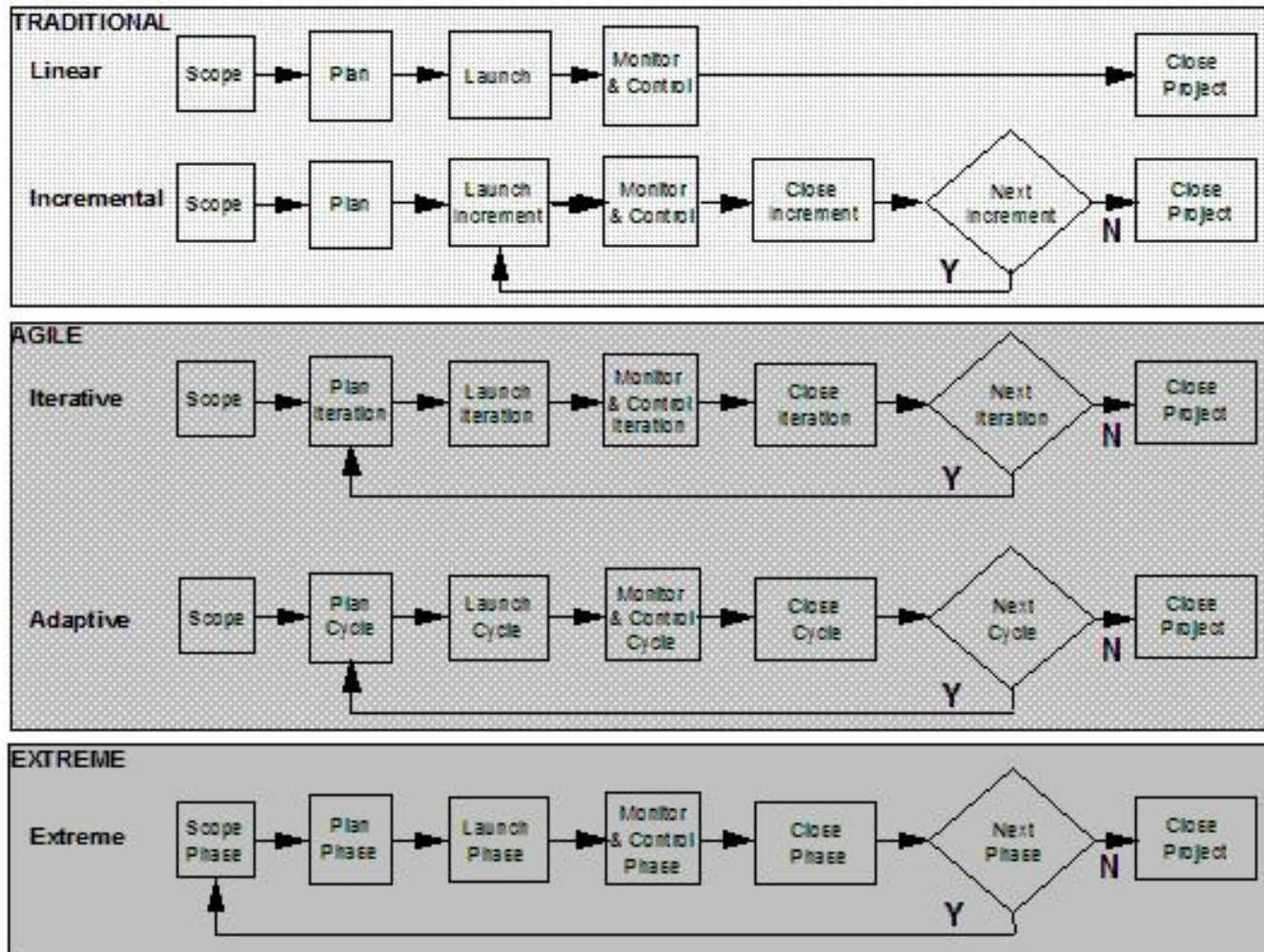
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Hybrid Project Management Approaches



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The 5 PMLC Models



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Recap of the Models

Similarities

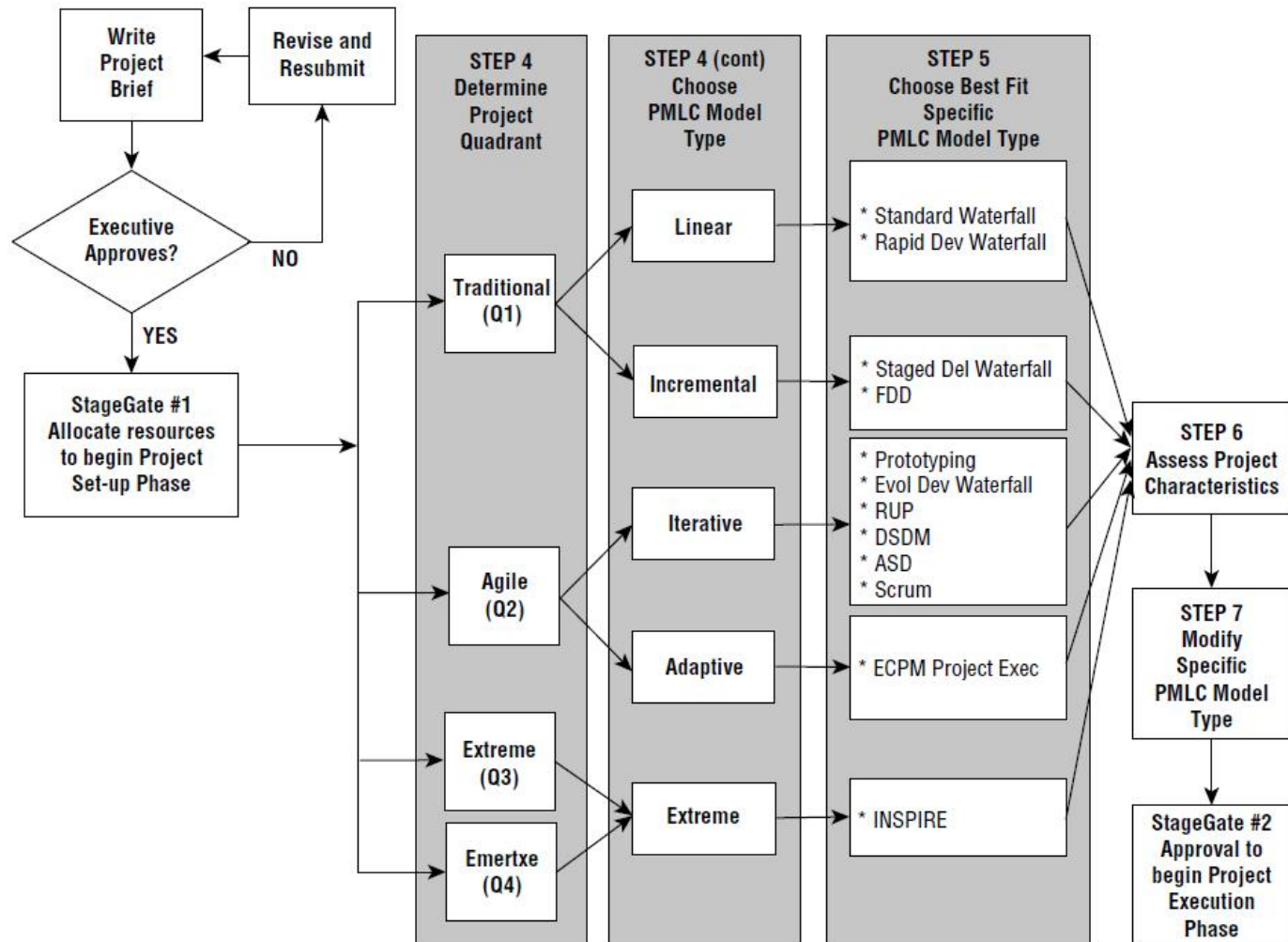
- All 5 Process Groups are used in each PMLC model
- Each PMLC model begins with a Scope Process Group
- Each PMLC model ends with a Close Process Group

Differences

- The models form a natural ordering (Linear, Incremental, Iterative, Adaptive, Extreme) by degree of solution uncertainty
- The processes that form repetitive groups recognize the effect of increasing uncertainty as you traverse the natural ordering
- Complete project planning is replaced by just-in-time project planning as the degree of uncertainty increases
- Risk management becomes more significant as degree of solution uncertainty increases
- The need for meaningful client involvement increases as degree of solution uncertainty increases

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Choosing the Best-Fit Model



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When to Use (1 of 2)

Linear

- Clearly defined solution and requirements
- Not many scope change requests
- Routine and repetitive projects
- Uses established templates

Incremental

- Same as linear but delivers business value early and often
- Some likelihood of scope change requests

Iterative

- Unstable or incomplete requirements and functionality
- Learn by doing and by discovery

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When to Use (2 of 2)

Adaptive

- Goal known but solution not known
- Solution highly influenced by expected changes
- New product development and process improvement projects

Extreme

- Goal and solution not known
- Through iteration converge on goal and solution
- Typically for R&D projects

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Exercises 2# in-class

■ Building a Mobile App for a Startup

Project Description:

A startup company wants to develop a mobile app that provides users with personalized workout plans and tracks their fitness progress. The app will include features such as exercise routines, meal planning, progress tracking, and social sharing. The startup aims to launch the app within six months to gain a competitive edge in the fitness market.

How to choose the suitable software management process for it?

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Additional Consideration

- Total Cost
- Duration
- Market Stability
- Technology
- Business Climate
- Number of Departments Affected
- Organizational Environment
- Team Skills and Competencies

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The KAs of Project Management

The Guide to the Project Management Body of Knowledge (PMBOK® Guide) and the Software Extension to the PMBOK® Guide (SWX).

- Project integration management,
- Project scope management,
- Project time management,
- Project cost management,
- Project quality management,
- Project human resource management,
- Project communications management,
- Project risk management,
- Project procurement management,
- Project stakeholder management.

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Project Management Process

Initiation and Scope Definition

- Determination and Negotiation of Requirements
- Feasibility Analysis
- Process for the Review and Revision of Requirements

An Example

Brightmouth College is a higher education institution which used to be managed by a local government authority but has now become autonomous. Its payroll is still administered by the local authority and pay slips and other output are produced in the local authority's computer centre. The authority now charges the college for this service. The college management are of the opinion that it would be cheaper to obtain an 'off-the-shelf' payroll package and do the payroll processing themselves.

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Project Management Process

Software Project Planning

- Process Planning
- Determine Deliverables
- Effort, Schedule, and Cost Estimation
- Resource Allocation
- Risk Management
- Quality Management
- Plan Management

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Project Management Process

Software Project Enactment

- Implementation of Plans
- Software Acquisition and Supplier Contract Management
- Implementation of Measurement Process
- Monitor Process
- Control Process
- Reporting

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Project Management Process

Review and Evaluation

- Determining Satisfaction of Requirements
- Reviewing and Evaluating Performance

An Example: How to judge whether this project is successful?

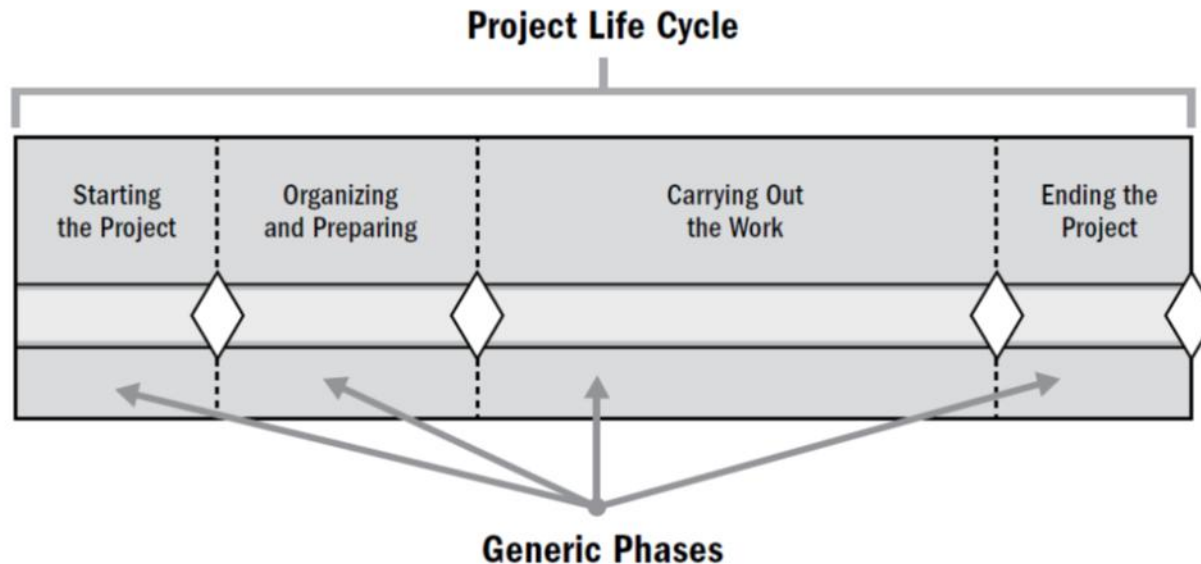
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Project Management Process

Closure

- Determining Closure
- Closure Activities



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Summary of Lesson Two

- Understanding the fundamentals of project management
- Challenges to effective project management
- Managing the creeps
- What are real requirements?
- Introducing project management life cycles
- The project landscape
- Traditional Project Management (TPM)
- Agile Project Management (APM)
- Extreme Project Management (xPM)
- Choosing the best-fit Model

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Assignment of Lesson Two

- Team construction & project choice
 - Design your course project, specify **Project's Name**.
 - The Deadline: Before **23:59** on **March 16th, 2026**
 - Team leader should submit the information in the form of replying to the homework at Canvas.